ENTITY	ATTRIBUTES	PRIMARY KEY
Department	dept_id, dept_name	dept_id
Student	student_id, matric_no, full_name, dept_id	student_id
Lecturer	lecturer_id, lecturer_name, dept_id	lecturer_id
Course	course_id, course_code, course_title, dept_id	course_id
Enrolment	enrolment_id, student_id, course_id, session	enrolment_id

1. Department → Student: One-to-Many (1:N). One department has many students, but each student belongs to one department.

2. Student → Course: Many-to-Many (M:N). One student can register for many courses, and one course can have many students. The Enrolment table resolves this relationship.

3. Lecturer → Course: Many-to-Many (M:N). A lecturer can teach many courses, and a course can have multiple lecturers across sessions. The Course_Lecturer table resolves this relationship.

First Normal Form (1NF)

matric_no	student_name	student_dept	course_code	course_title	lecturer	lecturer_dept	session
20231234	Ada Obi	Software Engineering	SEN307	Database Systems	Dr A. Okafor	Software Engineering	2026/2027
20231234	Ada Obi	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Engineering	2026/2027
20231235	John Musa	Computer Science	SEN307	Database Systems	Dr A. Okafor	Software Engineering	2026/2027
20231235	John Musa	Computer Science	COS301	Operating Systems	Mrs F. Eze	Computer Science	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Engineering	2026/2027

1NF was achieved by removing multiple values from individual cells and assigning each student-course registration its own row.

Second Normal Form (2NF)

Student

matric_no (PK)	student_name	student_dept
20231234	Ada Obi	Software Engineering
20231235	John Musa	Computer Science
20231236	Grace Okon	Software Engineering

Course

course_code (PK)	course_title
SEN307	Database Systems
SEN301	Software Architecture
COS301	Operating Systems

Enrolment

matric_no (FK)	course_code (FK)	session
20231234	SEN307	2026/2027
20231234	SEN301	2026/2027
20231235	SEN307	2026/2027
20231235	COS301	2026/2027
20231236	SEN301	2026/2027

2NF was achieved by removing partial dependencies and separating student information, course information, and enrolment records into different tables.

Third Normal Form (3NF)

Department

dept_id (PK)	dept_name
1	Software Engineering
2	Computer Science

Student

student_id (PK)	matric_no	full_name	dept_id (FK
1	20231234	Ada Obi	1
2	20231235	John Musa	2
3	20231236	Grace Okon	1

Lecturer

lecturer_id (PK)	lecturer_name	dept_id (FK)
1	Dr A. Okafor	1
2	Dr M. Bello	1
3	Mrs F. Eze	2

Course

course_id (PK)	course_code	course_title	dept_id (FK)
1	SEN307	Database Systems	1
2	SEN301	Software Architecture	1
3	COS301	Operating Systems	2

Enrolment

enrolment_id (PK)	student_id (FK)	course_id (FK)	session
1	1	1	2026/2027
2	1	2	2026/2027
3	2	1	2026/2027
4	2	3	2026/2027
5	3	2	2026/2027

Course_Lecturer

course_lecturer_id (PK)	course_id (FK)	lecturer_id (FK)	session
1	1	1	2026/2027
2	2	2	2026/2027
3	3	3	2026/2027